

[10] CacheBlend: Fast Large Language Model Serving for RAG with Cached Knowledge Fusion

- **Authors:** J. Yao, H. Li, Y. Liu, S. Ray, Y. Cheng, Q. Zhang, K. Du, S. Lu, & J. Jiang
 - **Venue & Year:** Proceedings of the Twentieth European Conference on Computer Systems (EuroSys '25), pp. 94–109
 - **Category:** Inference KV-Cache Optimization
 - **Associated Full Paper PDF:** [yao_2025_cacheblend.pdf](#)
-

1-Minute Executive Takeaway

Key Finding: Demonstrates that caching intermediate representations in RAG prevents GPU bottlenecks, reinforcing our pipeline caching philosophy.

Core Research Objective

To eliminate the redundant prompt prefill computation in RAG systems by dynamically reusing and fusing precomputed Key-Value (KV) cache states of retrieved document chunks.

Methodology & Architecture

Developed CacheBlend, an inference-engine extension that selectively merges KV-cache activations of retrieved chunks, reducing prompt token prefill overhead on GPU memory.

Key Empirical Findings & Evidence

Achieved 2.3× to 4.7× speedup in Time-to-First-Token (TTFT) and significantly reduced prefill compute for long-context RAG pipelines.

Direct Relevance to Our Kwiz Paper

Cited in Section 2.4 and Section 6.3. We distinguish our work from CacheBlend: whereas CacheBlend operates at the GPU inference layer (KV-cache fusion), Kwiz operates earlier in the pipeline (content hashing and embedding reuse during courseware indexing). We identify KV-cache fusion as the primary scaling path for future sub-second generation.